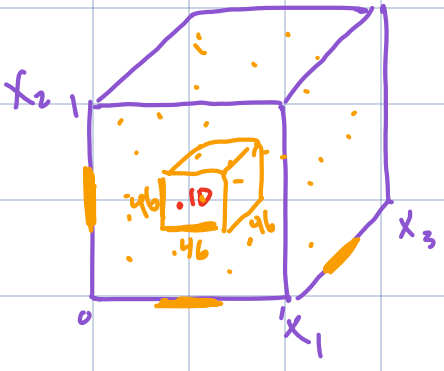
Agenda

- Curse of dimensionality
- Ridge regression
- Lasso regression

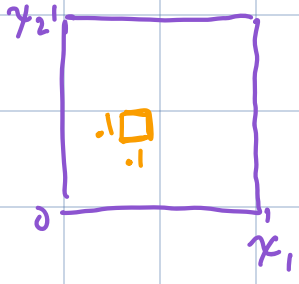
1. Curse of Dim



a "local" average of y over 10% of the data, we need 46% of the range of every predictor!

Problem 6 (b)

$p = 2$



e = edge length $= .1$

$.1^2 = .01$ ← volume

So 10% of range of X_1 and X_2 = 1% of data

6 (c)

$p = 100$ $\quad .1^{100} = 10^{-100}$

10% of range of every predictor = no training points

The fix: other ML models impose structure, e.g. a linear relationship

2. Regularized Linear Models

n: # of obs

p: # of predictors

$\beta = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}$ β_0 : intercept

$\mathbb{X}_{n \times p}$ (no column of 1s)

Recall AIC ($\lambda = 2$) penalty λ = # of $\hat{\beta}_j \neq 0$

BIC ($\lambda = \ln(n)$)

How about $\lambda \|\beta\|_1$ or $\lambda \|\beta\|_2^2$?

2.1 Regularization

Overfitting: model predicts much better on training set than on testing (unseen) data.

i.e. it doesn't generalize well.

Regularization

To improve generalization, add a penalty/constraint to the model's fitting object to discourage complexity

(reduce flexibility)

2.2 OLS problems

① If p is relatively large, variance is high

$$\frac{1}{n}\sum_{i=1}^{n} Var(\hat{y}_i) = (p+1)\frac{\sigma^2}{n}$$

② Multicollinearity

③ $p > n$ $X^T X$ is singular

3.1 Ridge regression / L2 Regularization

$$\hat{\beta}_\lambda^{ridge} = \underset{\beta_0,\beta}{\operatorname{argmin}} \left[\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right]$$

$$= \underset{\beta_0,\beta}{\operatorname{argmin}} \left[\sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j\right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right]$$

$\lambda \sum_{j=1}^{p} \beta_j^2$: $(L2\ Norm)^2$ $\quad \|\beta\|_2^2$

$$\hat{y}_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}$$

3.2 Standardize

We want all predictors on same scale,

so replace x_{ij} with

$$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j} \qquad s_j = \sqrt{\frac{1}{n}\sum_i (x_{ij} - \bar{x}_j)^2}$$

Why?

$X_1 \in \{0,1,2,3,4,5\}$ e.g. # of rooms

$X_2 \in [500, 5000]$ ℓ_2^2

← one will dominate the loss function.

$\beta_1 = 500$

$\beta_2 = 1$

Important: use $\bar{x}_j, s_j$ from training set

Lemma 3.2: If X is centered, then

$$\hat{\beta}_0 = \bar{y} \Rightarrow RSS = \sum_{i=1}^{n} \left((y_i - \bar{y}) - \sum_{j=1}^{p} x_{ij}\beta_j \right)$$

You can center y and get same $\hat{\beta}_s$ (w/out a β_0)

3.3 Solution for β

Aside: Hoerl + Kennard (1970) noticed

$(X^TX + \alpha I)$ fixes singularity so that

$(X^TX + \alpha I)^{-1}$ exists and behaves, so

$$\hat{\beta} = (X^T X + \alpha I)^{-1} X^T y$$ biased, but variance is lower!

$$RSS(\beta) = (y - X\beta)^T (y - X\beta) + \lambda \beta^T \beta$$

expand
derivative
set to $\emptyset$
solve for β

$$\hat{\beta}_\lambda^{ridge} = (X^T X + \lambda I)^{-1} X^T y$$

added some bias to $\hat{\beta}$s, but reducing variance.

4. Lasso / L1 Regularization

4.1 Defn

$$\hat{\beta}_\lambda^{lasso} = \underset{\beta_0, \beta}{\operatorname{argmin}} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

L1 Norm $\|\beta_1\|$

4.2 Standardize your predictors!

4.3 Solution for β

$$\frac{\partial}{\partial \beta} RSS(\beta) = -2X^T (y - X\beta) + \lambda \, sign(\beta)$$

undefined if $\beta = 0$

Must use numerical method, coordinate descent

4.4 Hard constraint

$$\hat{\beta}^{ridge} = \underset{\beta_0, \beta}{\operatorname{argmin}} \; RSS(\beta_0, \beta) \quad \text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 \le t$$

$$\hat{\beta}^{lasso} = \underset{\beta_0, \beta}{\operatorname{argmin}} \; RSS(\beta_0, \beta) \quad \text{subject to} \quad \sum_{j=1}^{p} |\beta_j| \le t$$

5. λ (tuning parameter)

- If $\lambda = 0 \Rightarrow$ OLS
- As $\lambda \nearrow$ $\hat{\beta}_j$s $\searrow$
- Lasso, shrinks $\hat{\beta}_j$s $\to 0$ (feature selection)
- Correlated features (x_1 and x_2): Ridge shrinks $\hat{\beta}_1 + \hat{\beta}_2$ together. Lasso shrinks one to 0.
- Choose λ using a test set, or use cross-validation

6. Summary

- $\hat{\beta}_0 = \bar{y}$
- Ridge will not shrink $\hat{\beta}_j$s $\to 0$
- Both add bias to $\hat{\beta}_j$, lower the variance,

and less flexible than OLS